

n. m. — the old surveyor's measure of the real.

Read the real.

Proof moves money.

The asset layer for AI —

resolve • read • verify • consent • attest • logbook

Why — make the physical world opposable.

How — unify the asset the way the world unified the person: from readings, not declarations.

A person is resolved a thousand times a day, across every system. An asset is never resolved — and that is what everyone decides on.

MISSION

\$50 trillion of assets — before counting real estate — are decided on from declarations. Our mission: every decision on a physical asset rests on a reading, not a claim.

The physical world was never made readable for AI.

\$50 T

of equipment, vehicles and products — before counting real estate — decided on from declarations, never from a reading (A1)

≈ 400 M

significant assets in the EU, ex-vehicles — each read from scratch at every function of every actor

10

identities per asset — CRM, field service, ERP, maintenance, warranty. None of them travels.

FIFTY YEARS OF IDENTITY INFRASTRUCTURE

Every wave solved a **who** — machine (DNS), person (OpenID, FIDO), consumer (cookie, IDFA, UID2), entity (LEI), device (passkeys), content (C2PA), agent (AP2). None solved a **what**. *This boiler* has no identifier, no attestation of state, no logbook.

A TRILLION TO KNOW A PERSON, NOTHING TO KNOW AN ASSET

Brands spend more than \$1 trillion a year on advertising, and tens of billions on the data stack that reconstructs — with varying certainty — what a person did. Nothing comparable exists to know what an asset is now: where it is, its state, who owns it, whether it can still be sold, insured, financed. **Every reading is paid for, used once, thrown away. Nothing accrues.**

There is no asset layer — and decisions are now taken by systems that never look.

Unify the asset the way the world unified the person — every significant asset keeps a logbook.

THE LAYER — SIX PRIMITIVES, ONE RAIL

01	Resolve	The permanent primitive: every act attaches a new answer to the same referent — where, state, owner, sellable, insurable. Deterministic, never probabilistic.
02	Read	Any device, on the asset — identity, attributes, state. Internal, free.
03	Verify	Does it satisfy rule X? Executed at capture, on-device, against a versioned ruleset. The billable act.
04	Consent	Signed yes or no — grantor, grantee, scope, purpose, validity, revocable.
05	Attest	Sealed, timestamped, logged — opposable, including against the reader.
06	Logbook	The asset's record: an index of attestations, never a copy of data. Accrues for life.

THE FUNCTIONS IT SERVES — THE JOBS DECIDERS PAY FOR

Settle	insurer · retailer · lessor — a claim, a return, a handback
Comply	aid payer · regulator — is it installed to code?
Operate	utility · maintainer · industry — the periodic check
Commit	landlord · lender — the state before the contract
Execute / pay	bank · payer — the milestone before the money
Consult	the AI agent — decides on an asset it never saw

Every decision resolves the asset again; the logbook is what the resolutions add up to — the asset graph, updated at every reading, under the holder's consent. The reader is whoever stands in front of the asset — customer, technician, inspector, agent. Then money moves — quote · underwrite · insure · finance · pay · sell · settle. The next decision starts from the logbook, not from scratch.

One engine reads the real

THE ENGINE

Proprietary tech that encodes any verification ruleset into any device — and runs it on the asset, on-device, in ~1 ms. In production with major European retailers.

CAPTURE INTEGRITY

The ruleset runs at capture, no network — never on uploaded files. Hardware attestation (Secure Enclave / StrongBox), authenticated position (Galileo OSNMA), attested path from street to floor. In the wedge the reader may be the party being checked — the record is opposable against them.

NOT ANOTHER MODEL

Deterministic compilation of a rule, not a world model. The rules belong to the client or the domain — public (ISO, DTU, CEE) or closed (a brand's policy). Arpent runs them; it never owns them.

RULES DON'T REWRITE FACTS

Each act carries the hash of the ruleset version it was read against. When the rule hardens in 2027, the 2025 act stays true against the 2025 rule. The logbook accrues; it never rewrites.

INTELLECTUAL PROPERTY

3–5 patent families on the mechanisms — capture integrity opposable to the reader, cryptographic ruleset versioning in the act, on-device compilation of arbitrary rulesets, attested capture sequence. Engine assigned to Arpent. Freedom-to-operate study before the raise. **The standard stays open (FRAND / royalty-free); the rail is what is defended.**

Priced like infrastructure — pay per verified outcome.

Product	Question answered	Price	Status
Read	What is it? — identity, attributes, state	not sold as an act	internal primitive — included in Verify plans
Verify	Does it satisfy rule X? — evidence + attestation	€2-20 / reading	the billable act — by depth: Resolve · Consent · Attest
Warranted (Phase 2)	Same act, with financial protection behind it	priced from risk, not compute	carrier-backed tier — appendix A10

WHO BUYS — THE DECIDER

The party whose function moves money on an asset: aid payer, insurer, lender, retailer, operator, landlord — and the agent. The decider sets who bears the fee: itself (control budget), the party being checked (like a vehicle inspection), or its control subcontractor.

WHO WIRES — THE FIELD SOFTWARE

API · CLI · MCP — one interface for tools and agents. Plugs into ServiceNow, Salesforce Field Service & industry clouds, Guidewire, Praxedo, ERP, maintenance tools. Each keeps its own copy; none can be the neutral referent between rivals' systems.

WHO READS — ANYONE WITH A PHONE

Customer, installer, inspector, agent. Capture is free and multiplies; what is paid is the sealing of the sequence. Degressive volume plans — never below €2, never a percentage. A billable act will never cost less than a booked appointment.

Fraud pays the first reading

The verify primitive is already in production — retail returns & warranty, closed rulesets, with IKEA, Fnac Darty and LVMH maisons. The first regulated wedge: **assisted-works programmes, Europe-wide — starting with two assets where the reading problem is beyond dispute.**

HEAT PUMPS

France mandates on-site checks — 100% by 2028. The visit stays; the payer pre-screens every site before submitting. One ruleset, one asset, one trigger.

EXTERNAL INSULATION (ITE)

The first fraud category of MaPrimeRénov' and CEE. Thickness, fixings, coverage: a reading no one contests. Same payers, same console.

France alone: ten payers · more than a million unique operations · €5–7 bn of aid a year. Fraud pays the first reading; the regulator supplies the ground truth — for free. **Supply side:** 300k installers reachable through one distributor, zero acquisition cost.

BEAT 1 · FRAUD

Payers already pay — control budgets of €150–400 per visit

BEAT 2 · LOGBOOK

Compliance draws on it; financing and insurance price off it; one reading, every entitlement

BEAT 3 · AGENTS

Pay for verified access — no proof of state, no settlement

The first programme is narrow by design — one asset class · one payer · one ruleset · one economic trigger. End-customer CAC = 0 · certifiers run on the rail · land on fraud, expand across functions.

Everyone reads. Nobody makes the reading opposable.

WHO READS

Vision models & remote inspection — Tractable, Bdeo, WeProov, generic VLMs; proptech — inspection & handover apps, site reality capture, lender draw inspections

WHO RECORDS

Captive registries — Verisk, Solera, CARFAX; DPP platforms; EAM systems

WHO ORCHESTRATES

Workflow software — ServiceNow, Salesforce Field Service, Guidewire, Praxedo

WHO CERTIFIES

Identity & liveness — iProov, eIDAS 2 wallets; QTSPs; inspection bodies

Readings, not acts: no capture integrity, no versioned rule, no seal. Channels, not rivals.

The registry serves the holding. One copy per system; ten identities per asset.

Workflows on top. Each keeps its own copy; none can be the neutral referent between rivals.

They verify who, or seal a file. We read what — and seal the reading at capture.

The real competitor is build-it-yourself. What no one else carries: the rule versioned in the act, the reading signed at the point of capture — opposable against the reader — the qualified tier (QTSP, eIDAS 2) that is accredited, not acquired, and a logbook nobody owns. *Regulation commoditizes reading — it cannot commoditize opposability.* The spec is free; operating its consequences is not: opposability carries liability, and liability doesn't fork.

The engine and the world

THE WORLD

Gregory Labrousse

Founder & CEO. 20 years making the real world machine-readable — nam.R, one of Europe's largest building-level data layers. Introduced Asset ID as a category at IIW, Copenhagen and GDC Geneva.

THE ENGINE

Bruno Antunes

Co-founder & CTO. On-device reading in ~1 ms — in production with major European retailers, on their own rulesets.

One founder read the market for two decades.

The other built the machine that reads it.

Now they meet.

Make the physical world opposable.

The wedge economics — and how big it gets

€2-20 ≈ 400 M

vs €150-400 — a ten-photo pre-screen against a human-control workflow; the customer must save >5× the fee

significant assets — EU, ex-vehicles (buildings · HVAC · home & collective systems)

PILOT KPIS — INSTRUMENTED DAY ONE

- cost saved / Arpent fee — target >5× (G1)
- false-pass rate — validated against mandated on-site controls (G2)
- human referral rate — declining, no hidden errors
- reading → opposable-act conversion
- functions per client per asset logbook (G4)

Floor	What it counts	Order of magnitude	Use
A · Acts	Verify acts sold, by cluster (11) and use case (37)	€13-23 bn/yr world · €1.8 bn EU bottom-up (matrix V3.3)	the TAM
B · Functions	Functions per client per asset logbook — the G4 metric	1.0 → 1.35 acceptances/op is worth more than any price move	the flywheel
C · Flows	Money that moves on a reading — aid, claims, loans, returns	≈ \$5,000 bn/yr	the ROI denominator
D · Stock	Assets decided on — equipment, vehicles, products, before real estate	> \$50,000 bn	why it matters — never a TAM

Three rungs, each priced honestly. 1 · Reuse — same logbook, next payer (north-star: second-payer acceptance without a new visit). 2 · Verticals — 37 use cases, 11 clusters, €1.8 bn EU. 3 · Geography — US insurance-first, \$1.2 bn ARR central, gated on G2. *The French wedge caps at €10-15M ARR by design — a rail never worth arguing about becomes mandatory by default. The wedge proves; the rungs pay.*

Own the asset layer.

Land on fraud — then expand across the functions of the same client, on the same asset logbooks. *Advance on proof, not on calendar.*

G 1 · U S E F U L N E S S

Customer saves >5× the fee — a PASS genuinely changes the workflow

G 2 · A C C U R A C Y

False-pass rate within risk budget — validated against mandated controls: the same accuracy every other function reuses

G 3 · I N S U R A B I L I T Y — T H E
T R I G G E R

A named carrier acknowledges the submission and opens technical underwriting — G3, not a date, triggers the warranted tier and the next raise

G 4 · E X P A N S I O N

Same client buys a 2nd function on the same logbooks — quoting · servicing · upsell: functions where nobody is being checked

What the seed proves: G1 + G2, with G3 set in motion — pilots → open spec → a carrier opens underwriting. Pilot metric — functions per client per asset logbook.

T H E A S K

The rail is built for agents; it is paid today by the parties who already decide on assets. Own the layer before the agent economy needs it — *every decision on a physical asset should rest on a reading, not a claim.*

How big is the substrate?

Reconciliation note — the slide-02 denominator is unchanged; the ladder below places it. Four floors, one ladder, never summed.

≈ 400 M	significant assets — EU, ex-vehicles	The wedge denominator (buildings, HVAC, home & grid systems).
≈ 750 M	one object = one asset — vehicles and smart meters included	The expansion universe (asset class N° 2 dial).
Billions	at product level	The universe the retail production already reads today.
> \$50,000 bn	before counting real estate — equipment & machinery ~\$50 T, vehicles ~\$15–20 T, consumer durables ~\$10–15 T; real estate (~\$380 T) is the container, not the object decided on [to be sourced]	Why it matters: assets are financed, insured, serviced and resold on declarations. Never a TAM, never the ROI denominator.

US — a price market, not a volume market. Insurance first.

Federal aided-works credits sunset end-2025 (25C/25D repealed, P.L. 119-21). What remains is better:

1 · STATE-MANDATED MITIGATION PRICING

CA, CO, LA, FL insurers must price verified mitigation, with a written right of appeal. FORTIFIED discounts run \$1,250–2,198/yr per home; Fannie/Freddie have already built the socket. One reading of the roof, every discount.

2 · UTILITY RATEPAYER DSM

\$8bn/yr, permanent, PUC-regulated QA budgets (\$80–240M field-verification pool); entry through implementers, as a signed-proof partner.

US pricing	1.5–2× per act
US TAM (central)	\$1.2bn ARR
Sequencing	Gated on G2 — advance on proof, not on calendar.
Opposability regime	signed evidence + ISO 17020 accreditation (ILAC/IAF mutual recognition) + contract — appendix A6

The deciders — who moves money on an asset, and why

Decider	Job (verb)	Pain today	Who can bear the fee	Cluster
Aid payer / regulator	Comply	Fraud; human control at €150–400; a failed lot freezes incentive payments	payer · or the installer as condition of submission · or the control call-centre	AT — the wedge
Insurer	Settle	Declared claim, late expertise, contested state	insurer · or the policyholder	AS
Lender / lessor	Commit · Execute	Collateral never seen; milestones declared; handback disputes	lender · or the borrower / lessee	BQ · VH
Retailer / brand	Settle · Execute	Returns, warranty, counterfeit — in production today	retailer	RD
Operator / maintainer	Operate	Periodic check redone from scratch; captive CMMS records	operator · or its inspection subcontractor	MM · UT · IN
Landlord / manager	Commit	Contested handover; regulatory state at letting	landlord · tenant (already the law for handovers)	IM
Installer / contractor	Prove & keep the relationship	Rejected files, delayed payment — and the logbook is the maintenance contract	installer (channel: 300k pros via one distributor)	supply side
AI agent	Consult	Decides on an asset it never saw	the agent's principal — no proof of state, no settlement	AG — multiplier

The « X for AI » map — and the empty cell

Category	Human primitive	Who names it	Known valuation / deal	Order of magnitude
Web / search for AI	search engine	Parallel, Exa, Firecrawl, Tavily	Parallel \$2bn (Series B \$100M, Apr 2026); Exa ~\$700M	50–100× ARR
Model gateway / routing	API gateway	OpenRouter	acquired by Stripe > \$7bn (Sep 2026)	strategic exit to the adjacent rail
Payments for AI	payment rail	Stripe (MPP, ACP), AP2, x402, Skyfire	Stripe \$159bn; Plaid \$6.1bn	~30× / ~15× revenue
Inference cloud	hosting	Together, Fireworks, Baseten	\$2–4bn each	30–60× ARR
Observability / evals	monitoring	Datadog, LangSmith, Braintrust	Datadog ~\$40bn; LangChain \$1.25bn	13–15× revenue
Security / guardrails	WAF	Lakera → Check Point; Protect AI → Palo Alto	~\$300M / ~\$600M exits	early M&A
Identity resolution (the mirror category)	people, for ads	LiveRamp, Epsilon (Publicis), Acxiom (Omnicom), Neustar (TransUnion), UID2	\$10–15bn market (2025) → \$20–30bn (2030); Epsilon \$4.4bn, Neustar \$3.1bn; LiveRamp ~\$813M rev.	probabilistic, ad-priced — ours is deterministic, decision-priced
Identity for AI agents	IAM	Okta, Affinidi, ToIP DTG, agentic FIDO	Affinidi: no valuation (Temasek)	—
Trust for AI	none	Cloudflare, C2PA, a dozen startups	dispersed	the cell to avoid
Physical AI	robotics	Nvidia, Physical Intelligence, Figure	PI \$5.6bn; Figure ~\$39bn	pre-revenue — the neighbour to watch
Asset for AI	cadastre · logbook · registry	nobody	—	the empty cell

Reading: « for AI » infrastructure prices at 50–100× ARR and exits early to the adjacent rail (OpenRouter → Stripe). Identity resolution built a \$15bn market unifying people for ads; nobody has unified the assets those people own — **asset resolution** is the budget line the decider will name, and the category Gartner can already write, mirror of a Magic Quadrant that exists.

What is protected, how — and what stays open

PATENTS — 3-5 FAMILIES ON MECHANISMS

Capture integrity opposable to the reader (rules executed at capture, signed against the party being checked) · cryptographic versioning of the ruleset inside the act (non-retroactivity) · on-device compilation of arbitrary rulesets (~1 ms, no network) · attested capture sequence (street fix → floor, barometer + inertial + session key). Stripe: 247 patents in 104 families — on mechanisms, never on « paying online ».

COPYRIGHT & TRADE SECRET — THE ENGINE

Engine assigned to Arpent; dated deposit of the code before the raise; explicit assignment contract with the CTO (who wrote what, when). Closed code, compartmented access — the regime Stripe and Plaid actually run on.

FREEDOM TO OPERATE — THE PLAID LESSON

Yodlee sued Plaid on seven patents (2014); Plaid settled by licensing Yodlee's 78-patent portfolio (2017). Captive registries and remote-inspection vendors hold twenty years of patents on geolocated photos, image measurement, signed reports. FTO study before the raise.

OPEN BY DESIGN — THE STANDARD

« Proof of asset-state » as a credential class carried into ToIP, UNTP, CEN — under FRAND / royalty-free licence. A rulebook Arpent held a patent on would be a silo with better presentation. The spec is free; operating its consequences is not.

The rail is global; presumption is a tier.

Regime	What makes it hold	Territory	What it costs the decider
1 • Legal presumption	Qualified seal, timestamp and electronic ledger (eIDAS 2) — presumption of integrity and origin; the burden of proof reverses	EU + EEA	lowest — the proof is presumed
2 • Accreditation	ISO 17020 inspection body accredited by COFRAC — signatory of the ILAC / IAF mutual-recognition arrangements: recognised by accreditation bodies in ~100 countries, MENA and LATAM included	worldwide	low — the reader is recognised
3 • Contractual / evidentiary	Signed evidence — JAdES / VC, hardware attestation, authenticated position — admissible in court, accepted by contract between parties and by the carrier. The DocuSign regime outside the EU	everywhere	priced by the carrier — the warranted tier

Several jurisdictions outside the EU run their own qualified regime (ICP-Brasil, UAE PASS, Switzerland): the act is sealed under the local regime by a local provider, the rail unchanged. The standards themselves — VC, DID, JWS, GS1, EPCIS, LEI, ISO 17020 — are global by construction; that is why they won. **A carrier that underwrites on signed evidence in the US and on qualified seals in Europe proves the value is in the act, not the regime.**

Nothing to reinvent — everything to use

Layer	Standard	What it gives the act	Arpent's use
Identifier	EN 18219 (May 2026) · W3C DID · GS1 Digital Link, SGTIN, GIAI · IEC 61406	A conformant, persistent, resolvable Asset ID at item level	Register issues (did:web on Arpent's domain); Resolve points to SGTIN, GIAI, RNB, DPP — never copies
Container & seal	ETSI JAdES (TS 119 182) · EN 18246 (ESDC, Sep 2026) · W3C VC 2.0 / SD-JWT	A structure signed by reference (sigD), long-term archive (B-LTA), qualified seal	The act is a JWS: at once an eIDAS qualified seal, a VC payload and an EN 18246 ESDC
Event	GS1 EPCIS (ISO/IEC 19987) · UNTP Traceability Event / Conformity Credential	What / when / where / why on an identified object; assessor + reference standard	« An EPCIS event made opposable » — the sixth UNTP profile
Position & device	Galileo OSNMA (Jul 2025) · Apple App Attest / Google Play Integrity · Secure Enclave / StrongBox	Authenticated signal, attested app, hardware-bound key	Position integrity — an attested path, not a GPS point
Presentation & consent	OID4VP / OID4VCI · DIDComm · FIDO PXP (offline) · DSP2 consent · FHIR Consent	Present to an insurer, a bank, an agent; consent with lifecycle	No protocol written; Consent = grantor, grantee, scope, purpose, validity, status
Issuer & trust	GLEIF LEI / vLEI · EU Trusted List · ToIP TRQP	Who signs, and whether they are entitled to	Emitter onboarding = KYB + role + rulesets allowed (the MID of the rail)
Birth of the object	EU DPP Registry (Jul 2026) · Battery passport (Feb 2027) · CPR DPP	A directory, not a data host — the EU just settled Arpent's architecture	The passport is the birth; the act is the life

Who keeps records on assets today — and for whom

Family	Examples	What they hold	Owner of the record	Relation to the rail
Captive registries	Verisk, Solera, CARFAX, Histovec	Claims history, vehicle history, property data	the holding	federated partners — the rail points, never copies
Product passports	EU DPP Registry, battery passport, Circularise, Spherity	What the product is at market placement	manufacturer / EU	identification anchor — birth, not life
Enterprise asset systems	IBM Maximo, SAP PM, Carl Source; GIAL-based fleets	Maintenance records of assets a company owns	the company	channel via Resolve (GIAL in); acts back to CMMS
Property & inspection tools	Homepad, Check & Visit, OpenSpace, Built	Handover reports, site progress, draw inspections	the workflow vendor	readers without seal — channels
Public anchors	Cadastre, RNB, BAN, VIN, HS codes (WCO)	Where and what, as coordinates	the state	pointed by Resolve — never a copy of registries
Commerce & identity graphs	Graas, Affinidi / DTG, Google Knowledge Graph	Product & customer graphs; persons, entities, agents as nodes	the platform	consumers of acts — they know the SKU before it ships, the issuer, the relation; never the state

State, lifecycle, value chain — three scales of one object

THE STATE — THE ATOM

One reading, at one instant, against one rule. It is the act: what is billed, what is patented, what is sealed.

THE LIFECYCLE — THE MOLECULE

The sequence of states. It is the logbook — the asset graph: every decision resolves the asset again, and the logbook is what the resolutions add up to. Aviation, vehicles and housing (CIL, 2022) already converged on the word.

THE VALUE CHAIN — THE MEDIUM

The actors who, at each moment of the lifecycle, decide and pay: manufacturer, installer, payer, insurer, lender, buyer, recycler. Traceability thinks in chains; identity thinks in nodes; nobody thinks in state.

TWO ECONOMIES ON ONE RAIL

Asset commerce — every exchange on a physical asset: sale, financing, insurance, lease, claim, retrofit, resale, recycling. The > \$50 T of floor D. The reading verifies the exchange — and triggers it: one reading of the old boiler tells a household every aid it is entitled to (KYC + KYAS = the entitlement, in one reading).

Asset support — every act in the life of the asset between two exchanges: installation, maintenance, inspection, repair. The logbook links the two: the installer who pays the first act buys the maintenance relationship — *the logbook is the customer relationship*.

A reading you can act on — and someone pays if it's wrong.

What is insured is not the asset — it is the reading. The decider acts on the act (pays the aid, releases the loan, settles the claim); if the act was wrong, the carrier reimburses up to a cap. Title insurance for the physical world: the proof, plus the money behind the proof. The carrier prices what it can measure — G2, the false-pass rate validated against the state's own controls, on millions of homogeneous acts. **Trigger: G3, a named carrier opens technical underwriting — not a date.**

THE FORMULA

Premium = Σ (amount acted upon on the act) $\times$ premium rate
Rate $\approx$ 0.5–2% of guaranteed amount (title insurance $\sim$ 0.5%, surety 1–3%)
Arpent = tier surcharge ($\sim 2 \times$ Verify) + originator / data share (10–20% of premium)

Rung	Amount acted upon	Premium pool (0.5–2%)	Arpent share
Wedge France — aided works	€5–7 bn/yr of aid	€25–140 M/yr	€5–25 M/yr — the size of the rail itself, no new client
Aided programmes, Europe	$\times 4\text{--}5$	€100–600 M/yr	€15–100 M/yr
Verticals — claims, collateral, handbacks, returns	1–5% of the $\sim$ \$5,000 bn/yr of floor C flows, at term	\$250 M – \$5 bn/yr	\$40 M – \$1 bn/yr — gated on G4
Floor D — the stock	\$50 T	never a premium base	—